

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

Instructor: Ijaz Ul Haq, Ph.D.

ijazhaq@umich.edu

Location: MSB 104



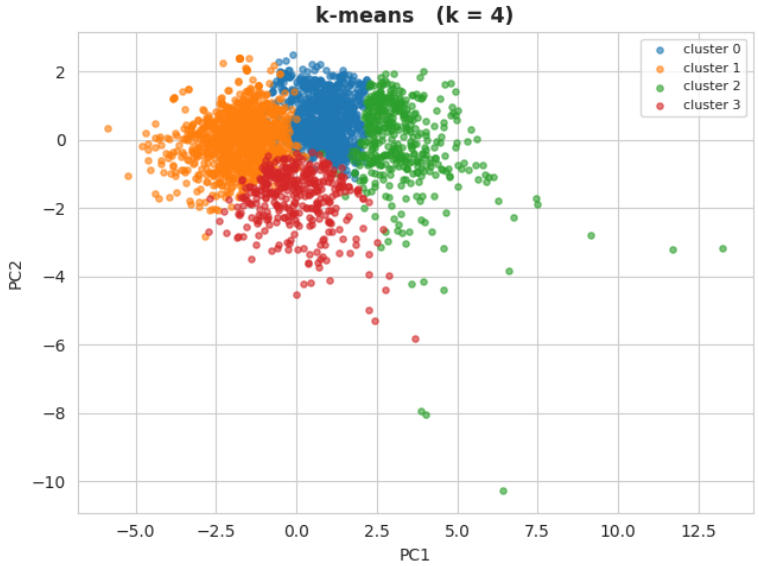
Lecture 10
Cluster Validation and Interpretation

Today's class:

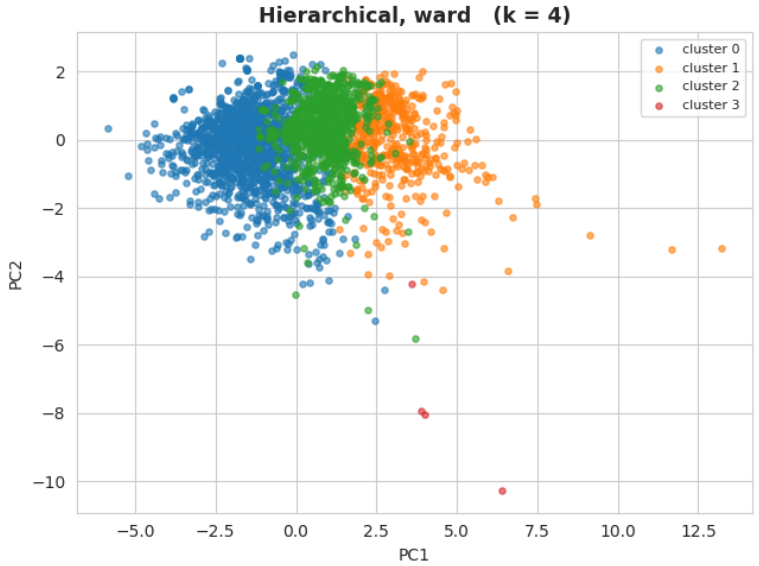
- ***Cluster Validation***
- ***Silhouette***
- ***Davies-Bouldin***
- ***Agreement across seeds***
- ***Adjusted Rand Indices (ARI)***

Where We Left Off

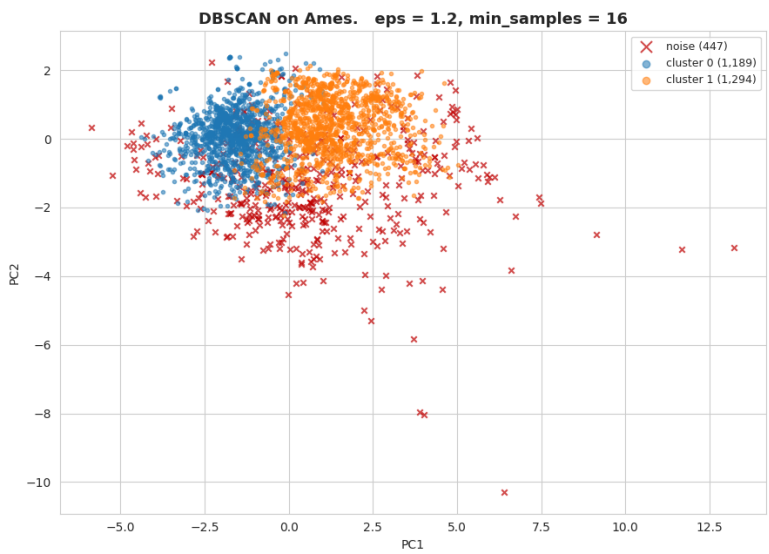
k-mean



Hierarchical



DBSCAN

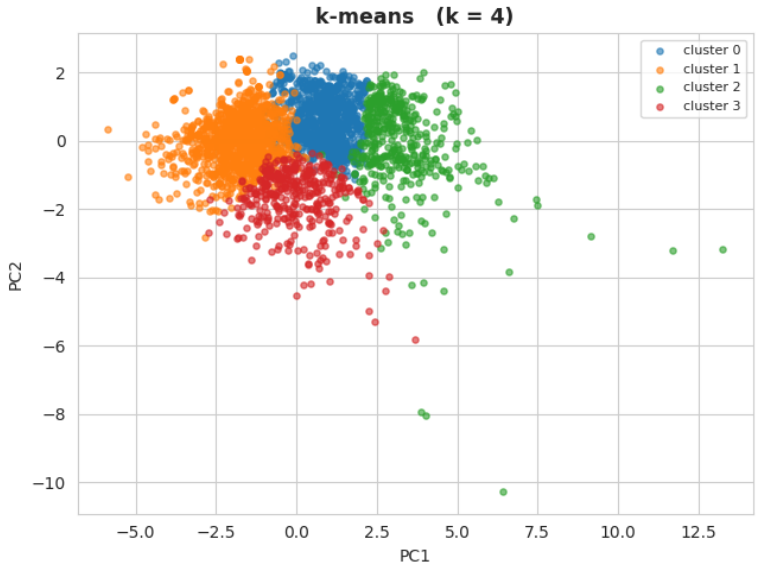


The assessor has no price bands yet. Given everything we know about these 2,930 houses, do they fall into natural groups? And if so, what are those groups?

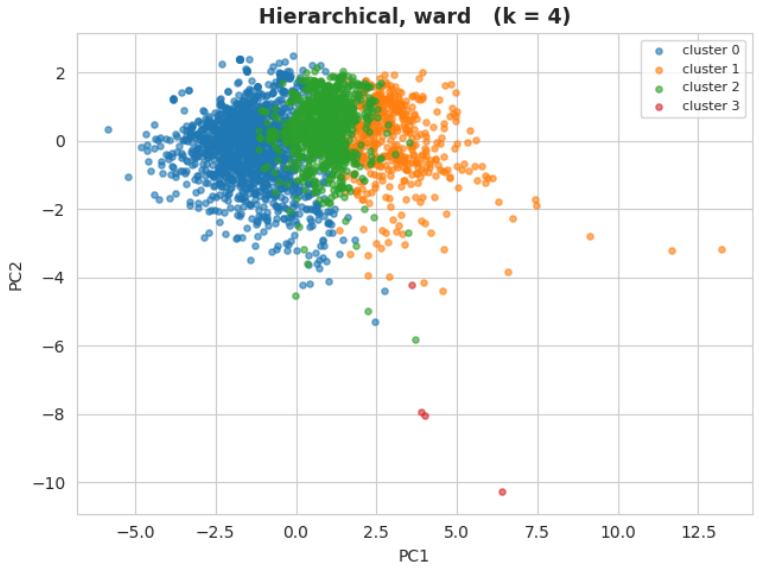
- **k-means:** four clusters, every house in one of them, boundaries are straight lines.
- **Hierarchical:** four clusters, different boundaries, same houses.
- **DBSCAN:** two clusters plus 447 houses it refused to place.

Where We Left Off

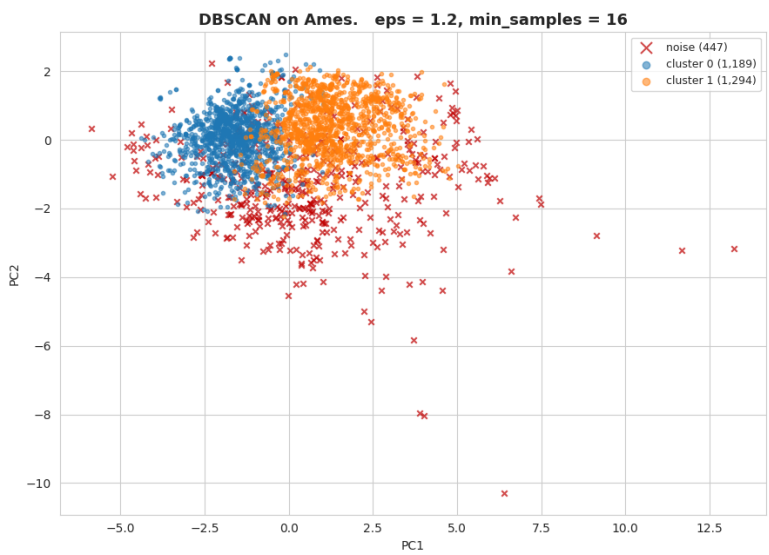
k-mean



Hierarchical



DBSCAN



- You cannot tell by looking. All three look plausible.
- There is no accuracy to compute. Nobody labelled these houses.
- Each method was obeying its own rule, and each rule gave a different result.

Cluster Validation

The problem

- No labels, so no accuracy, no precision, no confusion matrix.
- Every algorithm returned clusters. None of them can be wrong.
- Looking at the plot tells you nothing. All three looked plausible.

What Validation Actually Measures

- **Are the clusters separated?** Is there real space between them, or did we cut one blob into pieces?
- **Are the clusters tight?** Are the houses inside a cluster actually similar to each other?
- **Is the structure stable?** Change the seed or the sample. Do the same groups come back?
- **Is it better than nothing?** Cluster pure random numbers. Does your result score any better?

Kind	What it uses	Example
Internal	The data only. No labels needed	Silhouette, Davies-Bouldin
Stability	Repeated runs on the same data	Agreement across seeds
External	Labels, on the rare occasion you have them	ARI, NMI

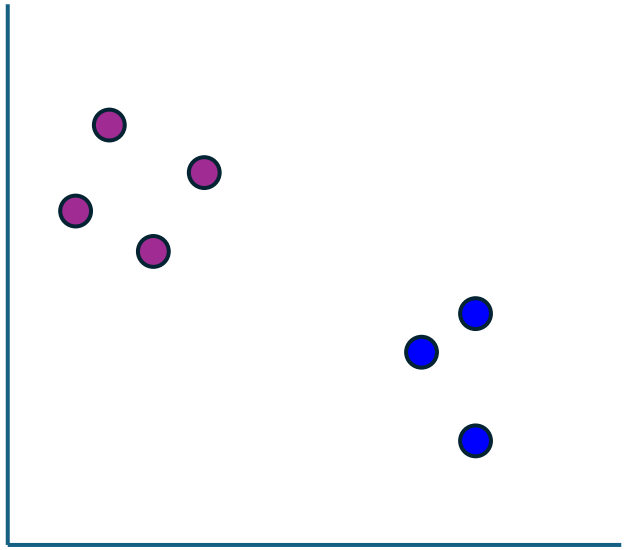
None of these tells you the clusters are correct.
They tell you whether the clusters are worth believing.

Cluster Validation

Silhouette

The silhouette coefficient combines ***cohesion*** and ***separation*** for an individual point: it compares the point's average distance to other points in its own cluster against its average distance to points in the nearest other cluster, yielding a value between −1 and 1, where values near 1 are best.

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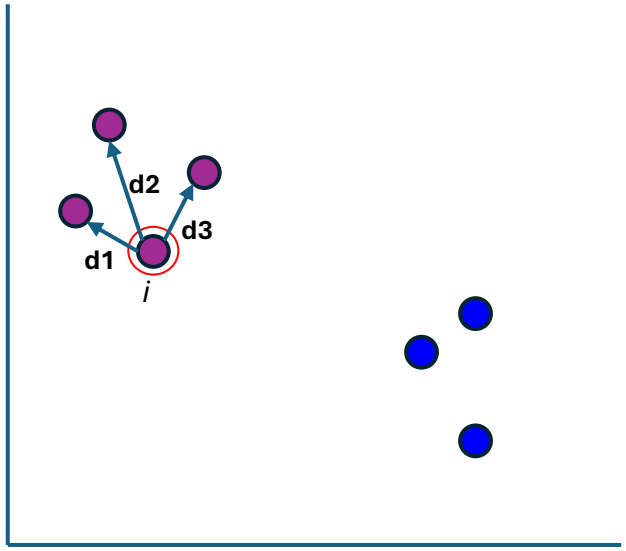


Cluster Validation

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Cohesion: *similarity within cluster*
$a_{(i)}$ = mean of all the distances within the cluster

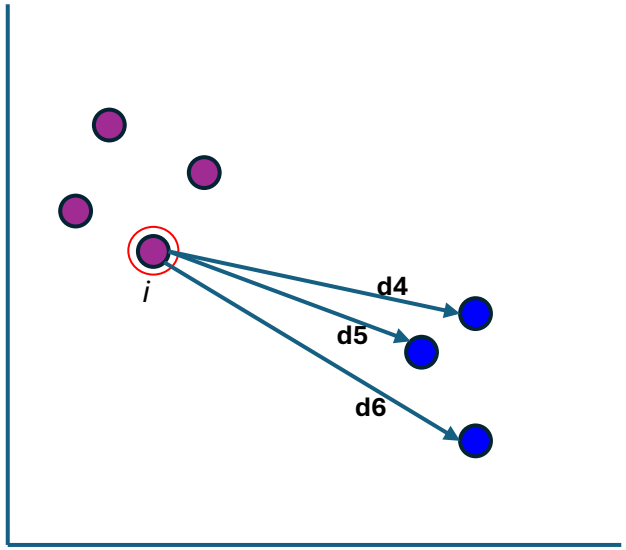
Separation: *difference from other clusters*

Cluster Validation

Silhouette

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Cohesion: *similarity within cluster*

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Separation: *difference from other clusters*

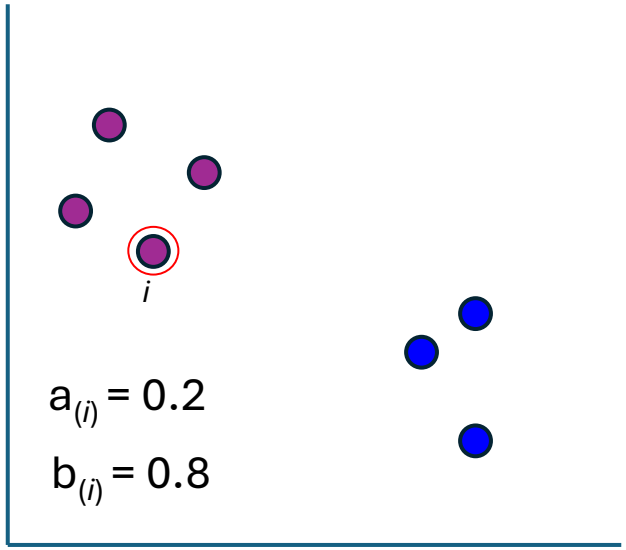
$b_{(i)}$ = mean of all the distances in other cluster

Cluster Validation

Silhouette

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silhouette coefficient: $\frac{b - a}{\max(b, a)} = \frac{0.8 - 0.2}{\max(0.8, 0.2)} = \frac{0.6}{0.8} = 0.75$

Cohesion: *similarity within cluster*

$a_{(i)}$ = mean of all the distances within the cluster

Separation: *difference from other clusters*

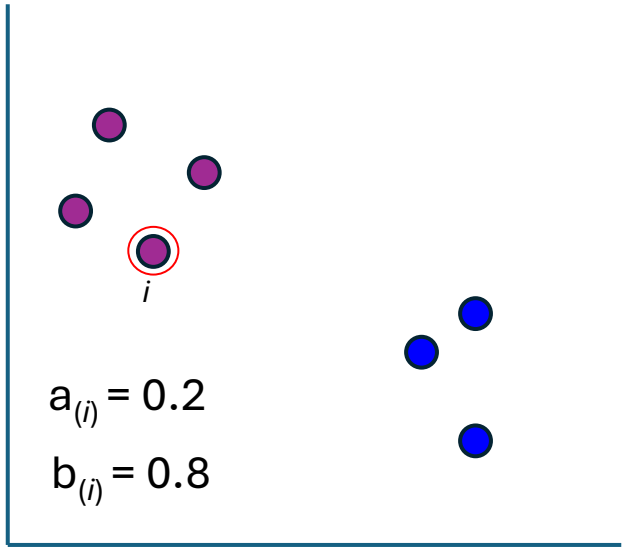
$b_{(i)}$ = mean of all the distances in other cluster

Cluster Validation

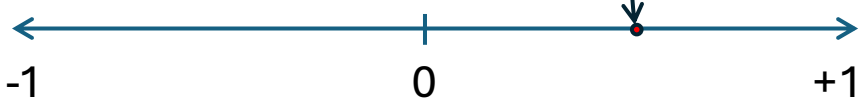
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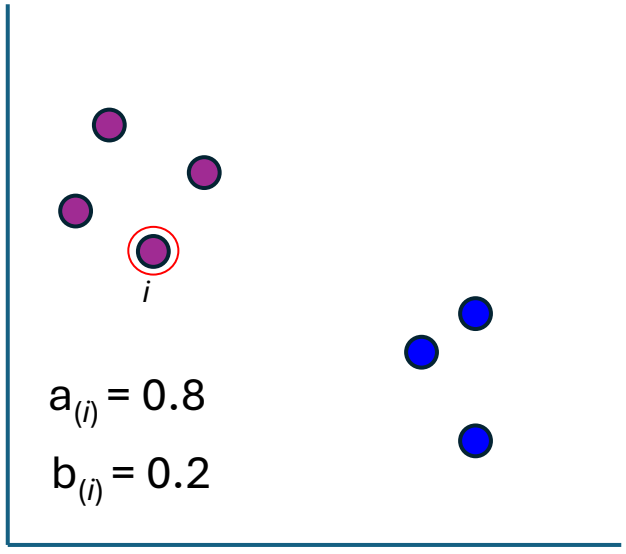
+1	Deep inside my cluster, far from any other
0	Sitting on the boundary. Could belong to either
−1	Closer to another cluster than to my own

Cluster Validation

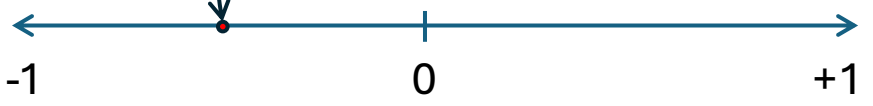
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silhouette coefficient: $\frac{b - a}{\max(b, a)} = \frac{0.2 - 0.8}{\max(0.2, 0.8)} = \frac{-0.6}{0.8} = -0.75$



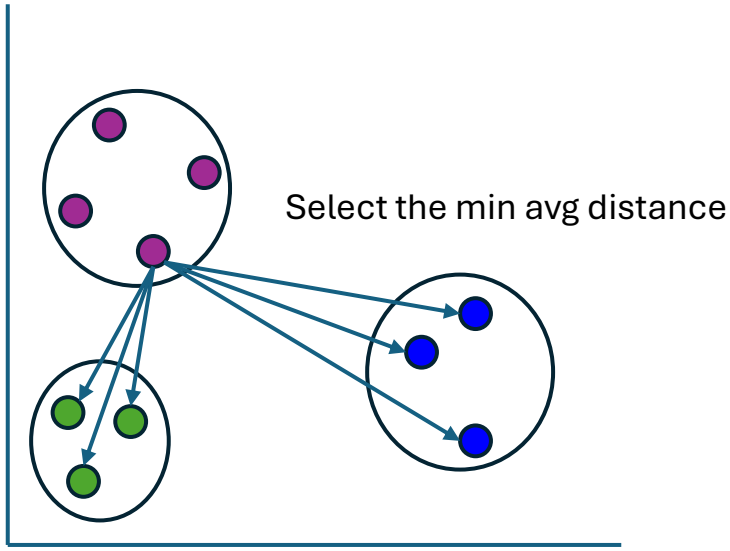
+1	Deep inside my cluster, far from any other
0	Sitting on the boundary. Could belong to either
−1	Closer to another cluster than to my own

Cluster Validation

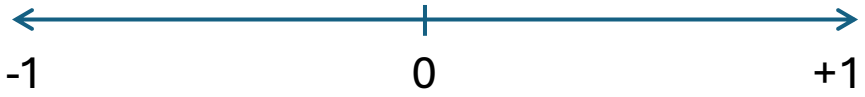
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silhouette coefficient: $\frac{b - a}{\max(b, a)}$



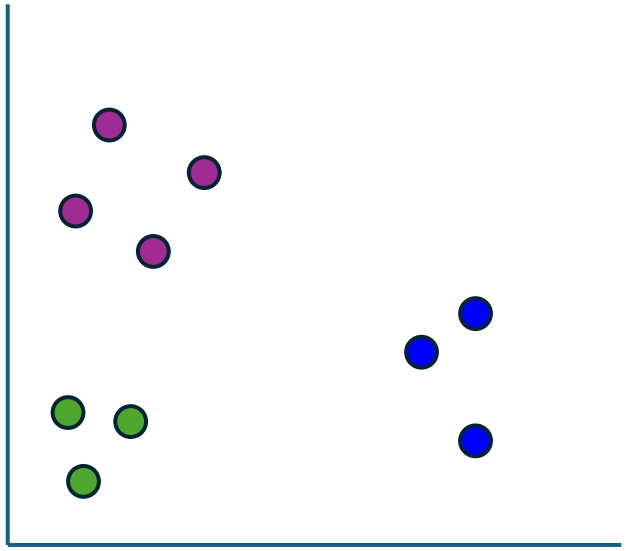
+1	Deep inside my cluster, far from any other
0	Sitting on the boundary. Could belong to either
−1	Closer to another cluster than to my own

Cluster Validation

Davies-Bouldin

The Davies-Bouldin index measures, for each cluster, how similar it is to its most ***similar neighbor***, by comparing their ***combined internal spread*** against the distance between them, then averages that worst case across all clusters.

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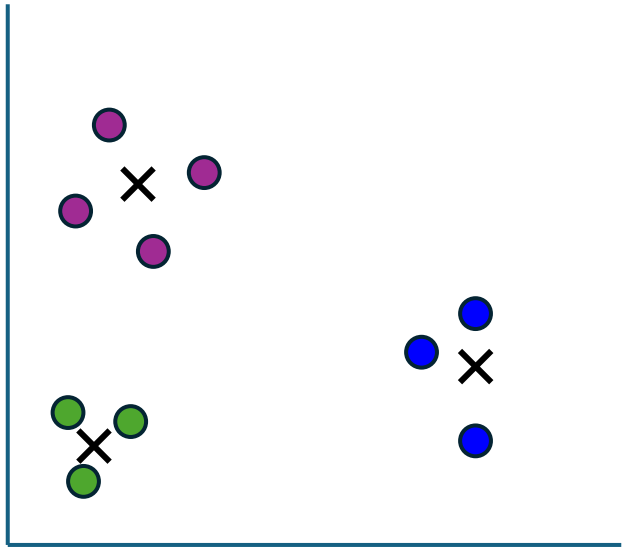
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Step 1: find the centroid of each cluster

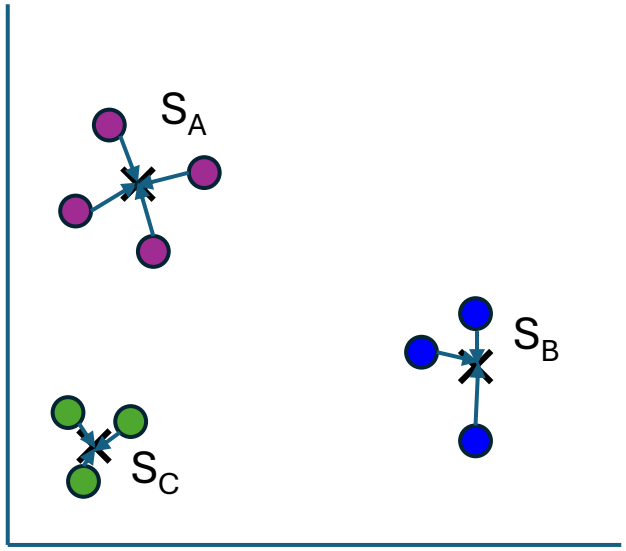


Cluster Validation

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Step 2: measure the scatter (S) of the clusters
S= mean distance of points to their centroid

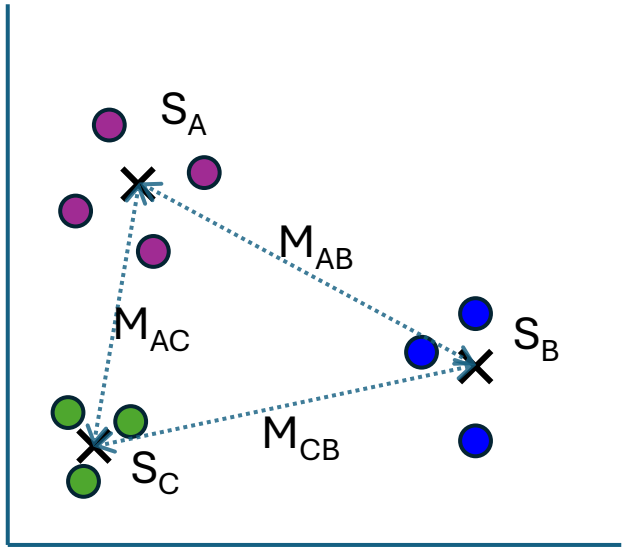
Cluster Validation

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Step 3: measure the distance between each pair centroids (M)

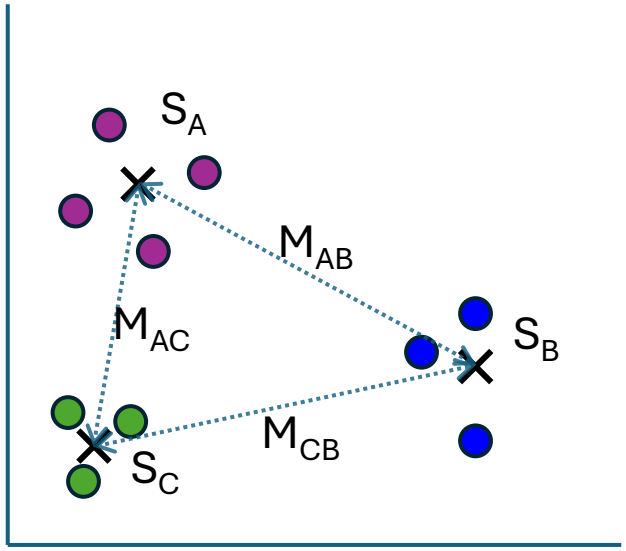


Cluster Validation

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Step 4: compute the ration (R)

DB Ratio (R)= $\frac{S_A + S_B}{M_{AB}}$

R_{AB} = 0.44

R_{AC} = 0.90

R_{BC} = 0.33

S_A = 1.0

S_B = 1.2

S_C = 0.8

M_{AB} = 5.0

M_{AC} = 2.0

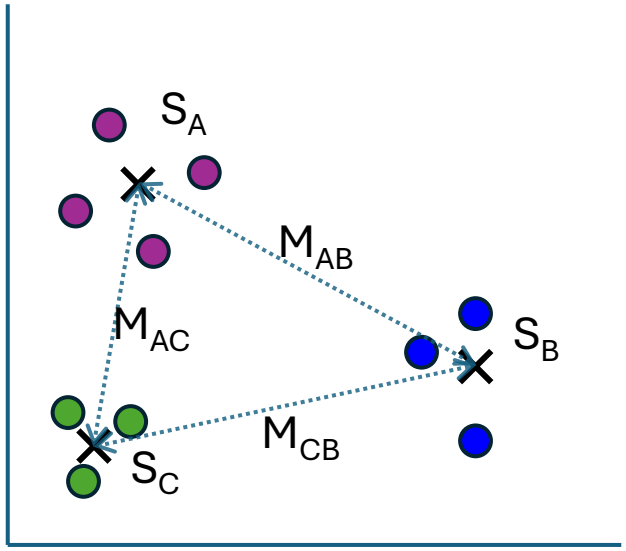
M_{CB} = 6.0

Cluster Validation

Davies-Bouldin

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Step 5: select the worst pair for each cluster

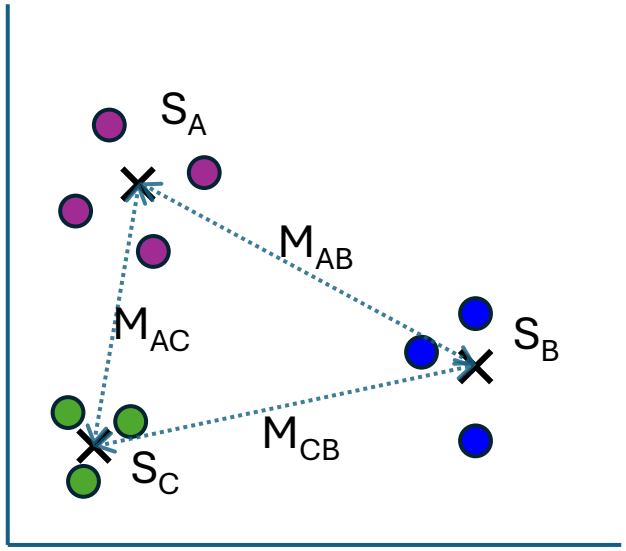
$R_{AB} = 0.44$	A: [0.44, 0.90]	0.90
$R_{AC} = 0.90$	B: [0.44, 0.33]	0.44
$R_{BC} = 0.33$	C: [0.90, 0.33]	0.90

Cluster Validation

Davies-Bouldin

The Davies-Bouldin index measures, for each cluster, how similar it is to its most ***similar neighbor***, by comparing their ***combined internal spread*** against the distance between them, then averages that worst case across all clusters.

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Step 6: take an average of the worst R of each cluster

$R_{AB} = 0.44$	A: [0.44, 0.90]	0.90
$R_{AC} = 0.90$	B: [0.44, 0.33]	0.44
$R_{BC} = 0.33$	C: [0.90, 0.33]	0.90

$$R = \frac{0.90 + 0.44 + 0.90}{3}$$

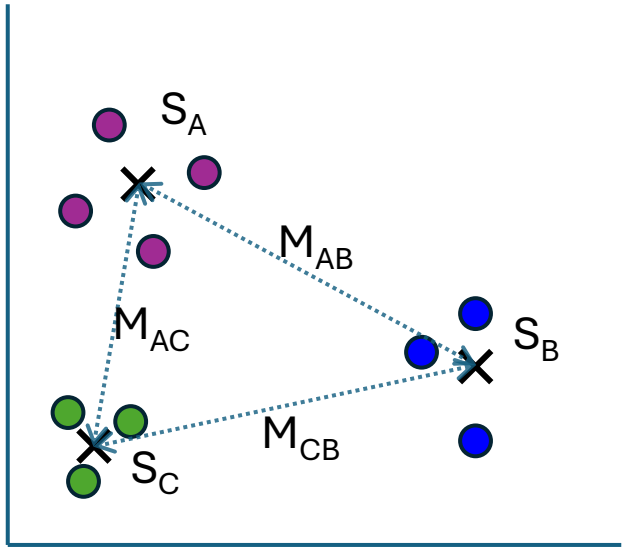
$$R = 0.75$$

Cluster Validation

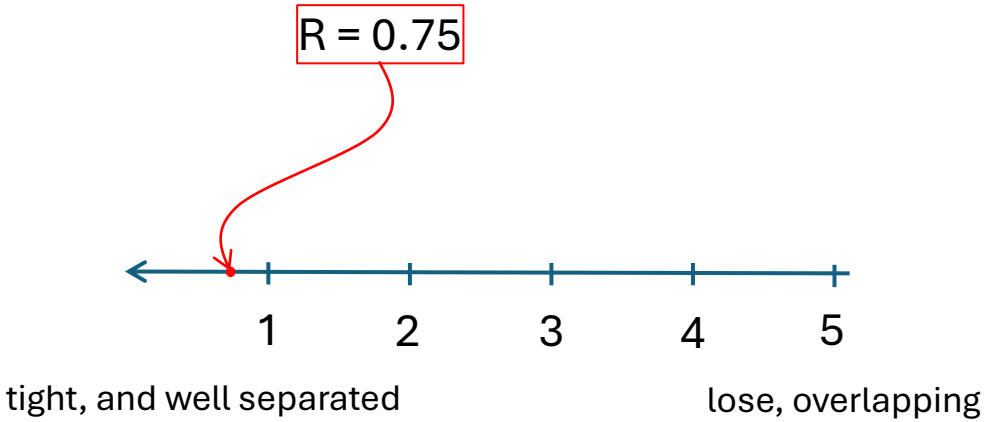
Davies-Bouldin

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Step 6: take an average of the worst R of each cluster



Cluster Validation

Agreement Across Seeds

The Problem

- k-means starts from random centroid positions.
- Run it again with a different random start and you may get a different answer.
- Silhouette tells you whether a clustering is tight. It does **not** tell you whether you would get it again.

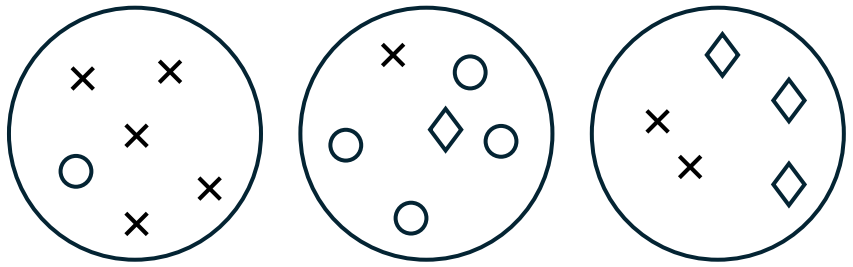
The Test

- Run k-means several times, changing only the random seed.
- Compare each run against the first.
- If the structure is real, the same groups should come back every time.

Cluster Validation

Adjusted Rand Indices (ARI)

The adjusted Rand index measures the agreement between two clusters, counting pairs of points rather than labels, and corrected for the agreement expected by chance, so that random clusteris score 0 and identical clusters score 1.



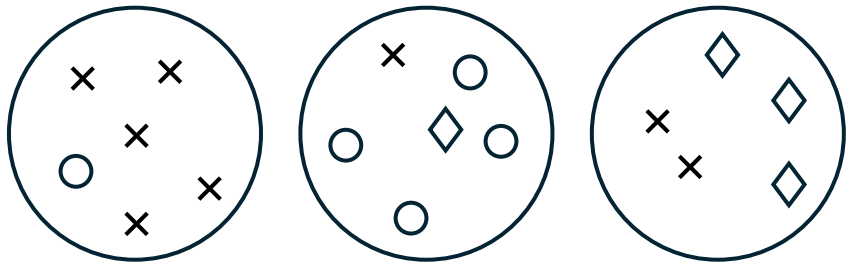
	×	○	◇	Total
Cluster 1	5	1	0	6
Cluster 2	1	4	1	6
Cluster 3	2	0	3	5
Total	8	5	4	17

Step 1: Build the contingency table
Count how many of each shape landed in each cluster.

Cluster Validation

Adjusted Rand Indices (ARI)

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	×	○	◇	Total
Cluster 1	5	1	0	6
Cluster 2	1	4	1	6
Cluster 3	2	0	3	5
Total	8	5	4	17

Step 2: count total number of pairs

n=17 $^{n}C_{2}$ $n(n-1)/2$

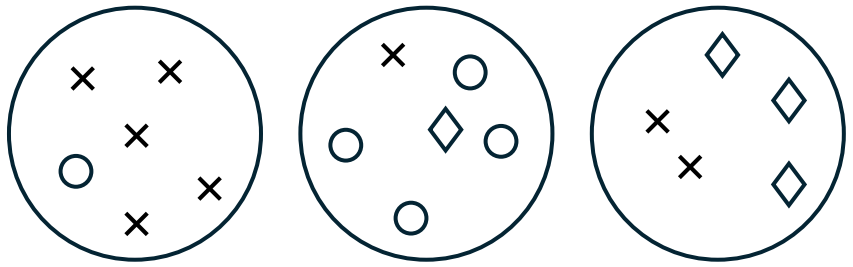
$17(17-1)/2$

$17(16)/2 = 136$

Cluster Validation

Adjusted Rand Indices (ARI)

The adjusted Rand index measures the agreement between two clusters, counting pairs of points rather than labels, and corrected for the agreement expected by chance, so that random clusteris score 0 and identical clusters score 1.



	×	○	◇	Total
Cluster 1	5	1	0	6
Cluster 2	1	4	1	6
Cluster 3	2	0	3	5
Total	8	5	4	17

Step 3: Pairs the two agree on (the cells)

$$= \binom{5}{2} + \binom{4}{2} + \binom{2}{2} + \binom{3}{2}$$

$$= 10 + 6 + 1 + 3$$

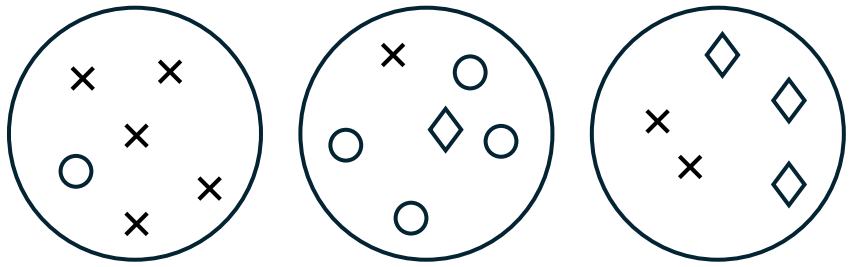
$$A = 20$$

Cluster Validation

Adjusted Rand Indices (ARI)

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A = 20



	×	○	◇	Total
Cluster 1	5	1	0	6
Cluster 2	1	4	1	6
Cluster 3	2	0	3	5
Total	8	5	4	17

Step 4: Pairs inside each cluster (the rows)

$$= \binom{6}{2} + \binom{6}{2} + \binom{5}{2}$$

$$= 15 + 15 + 10$$

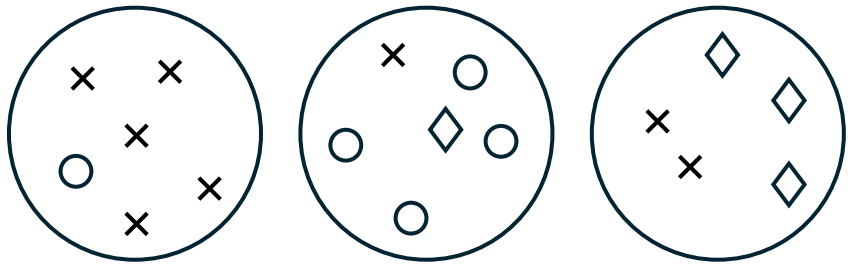
B = 40

Cluster Validation

Adjusted Rand Indices (ARI)

The adjusted Rand index measures the agreement between two clusters, counting pairs of points rather than labels, and corrected for the agreement expected by chance, so that random clusteris score 0 and identical clusters score 1.

A = 20 B = 40



	×	○	◇	Total
Cluster 1	5	1	0	6
Cluster 2	1	4	1	6
Cluster 3	2	0	3	5
Total	8	5	4	17

Step 5: Pairs inside each shape (the columns)

$$= \binom{8}{2} + \binom{5}{2} + \binom{4}{2}$$

$$= 28 + 10 + 6$$

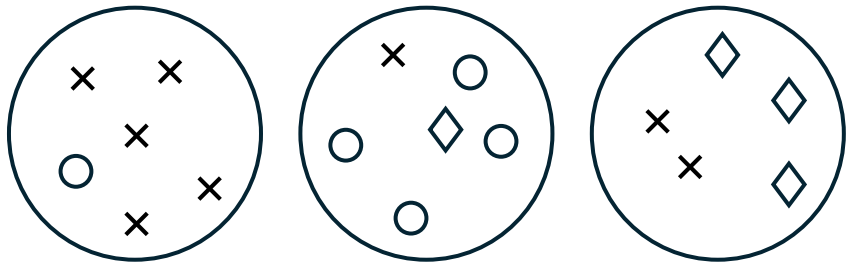
C = 44

Cluster Validation

Adjusted Rand Indices (ARI)

The adjusted Rand index measures the agreement between two clusters, counting pairs of points rather than labels, and corrected for the agreement expected by chance, so that random clusteris score 0 and identical clusters score 1.

A = 20 B = 40 C = 44



	×	○	◇	Total
Cluster 1	5	1	0	6
Cluster 2	1	4	1	6
Cluster 3	2	0	3	5
Total	8	5	4	17

Step 6: What you would expect by chance

Expected = (B × C) / total pairs

= (40 × 44) / 136

= 1760 / 136

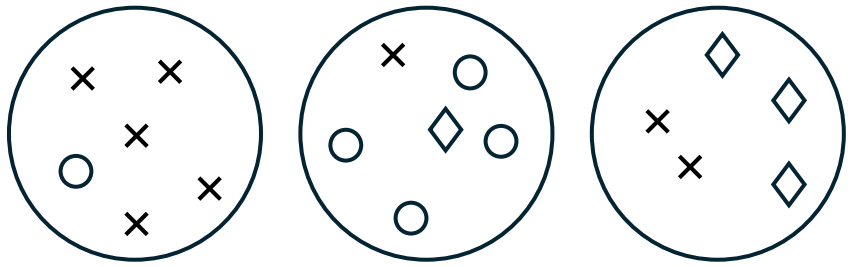
= 12.94

Cluster Validation

Adjusted Rand Indices (ARI)

The adjusted Rand index measures the agreement between two clusters, counting pairs of points rather than labels, and corrected for the agreement expected by chance, so that random clusteris score 0 and identical clusters score 1.

A = 20 B = 40 C = 44



	×	○	◇	Total
Cluster 1	5	1	0	6
Cluster 2	1	4	1	6
Cluster 3	2	0	3	5
Total	8	5	4	17

Step 7: The maximum possible

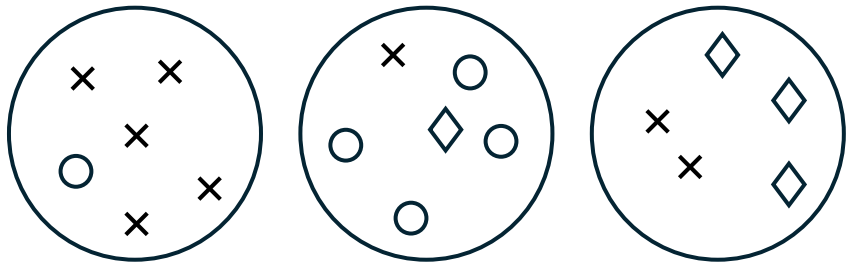
Max = (B + C) / 2
= (40 + 44) / 2
= 42

Cluster Validation

Adjusted Rand Indices (ARI)

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	×	○	◇	Total
Cluster 1	5	1	0	6
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Total	8	5	4	17

Step 8: put all together

ARI = (A – Expected) / (Max – Expected)
= (20 – 12.94) / (42 – 12.94)
= 7.06 / 29.06
= 0.24

ARI = 0.24. Poor agreement.

- The clustering got **20** same-shape pairs right.
- Random guessing would have got **13**.
- A perfect clustering would have got **42**.